

Trained



train



Large labelled training set

Environment-specific features

A

Standard ✓

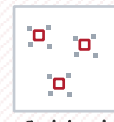
Cross-env. ✗

Training-free, prior work



Frozen patch features

→



Codebook



Segmentation CLS attention

Prior-selected / aggregated features

B

Standard ✗

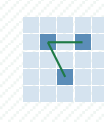
Cross-env. —

Ours

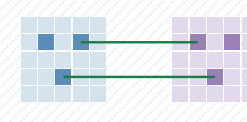


Frozen patch features

→



Within-image
Balanced, matchable patch evidence



Across-image
Balanced, matchable patch evidence

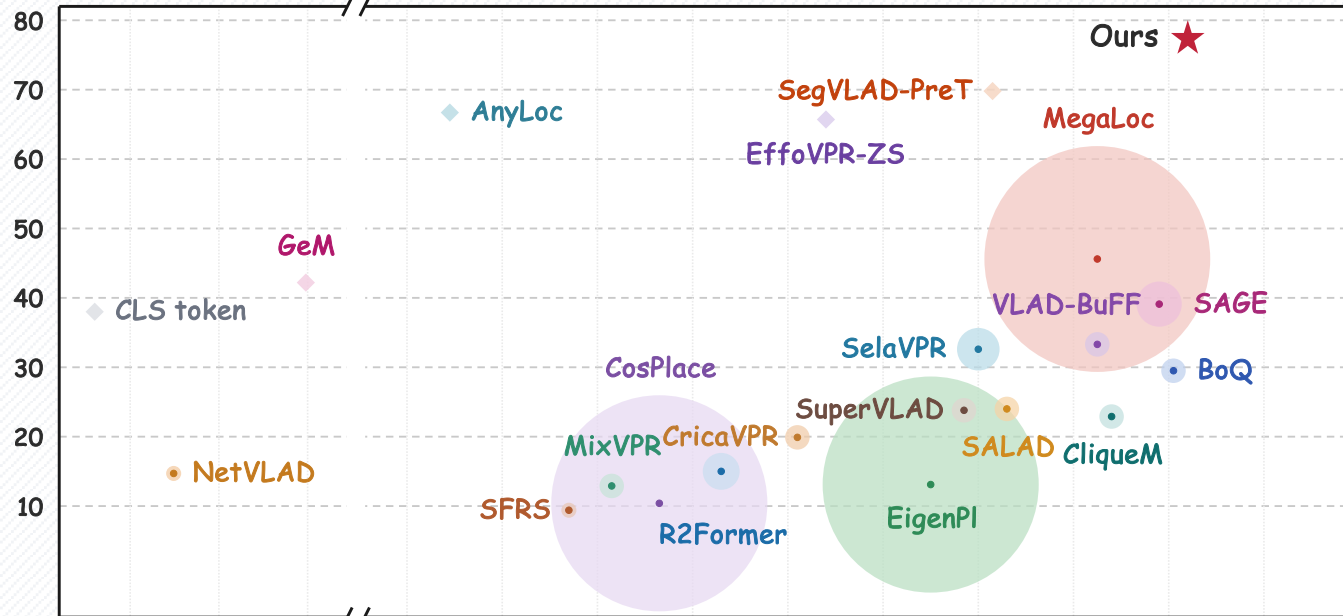
C

Standard ✓

Cross-env. ✓

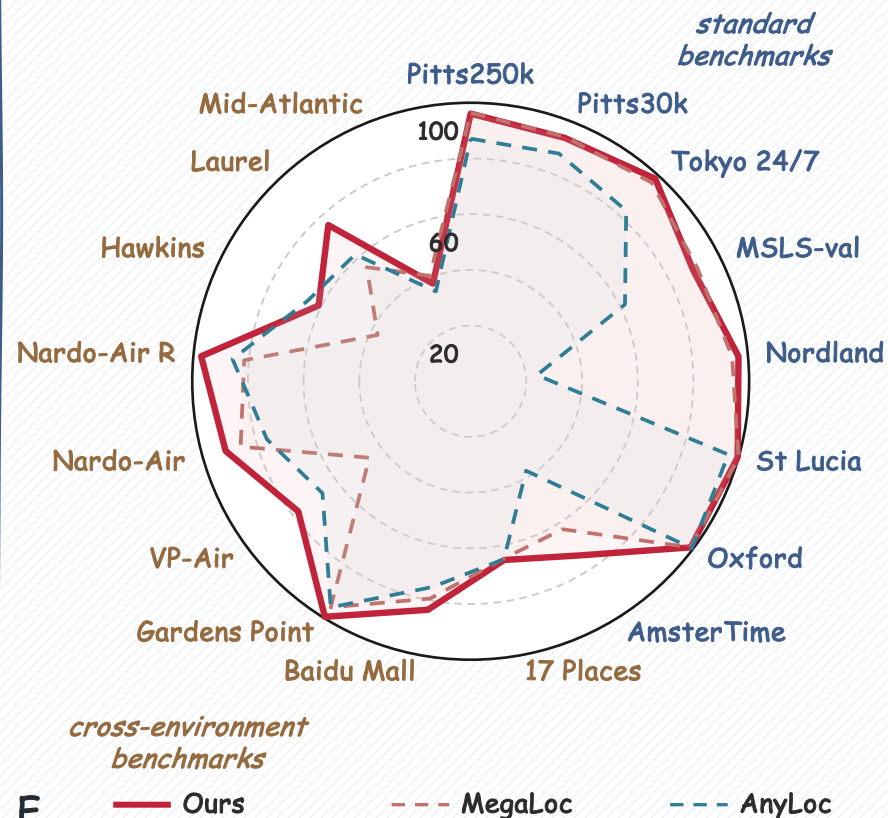
Training cost vs. performance

VP-Air R@1 (cross-environment benchmark, %)



Tokyo 24/7 R@1 (standard benchmark, %)

Performance on 17 benchmarks



cross-environment benchmarks

E

Ours

MegaLoc

AnyLoc

D

★ ours (training-free)

♦ training-free

● trained

● 1 million images

● 5 million images